

# Noisy VQE Optimization and Backend-Calibrated Noise

## Removing two idealizations from the Block-4 noise study

Seminar notes

July 4, 2026

### Abstract

The Block-4 noise study so far made two convenient assumptions: the variational state was optimized on a *noiseless* landscape (noise applied only at evaluation), and the noise itself was a *hand-set uniform* model (one per-cx depolarizing rate, or one global  $T_1/T_2$ ). This note documents lifting both. First, we put the noise *inside* the optimizer, so every cost evaluation is a noisy density-matrix simulation. The finding is a clean dichotomy: for **unital** noise (depolarizing) noise-aware optimization changes nothing – which retroactively *validates* the noiseless-preparation idealization – while for **non-unital** noise (amplitude damping) it raises the *measured* edge string but *lowers* the noiseless fidelity to the true ground state. Minimizing the device-level cost is therefore not the same as preparing the state: a better number can be a worse state. Second, we replace the toy noise with `NoiseModel.from_backend` on a recorded IBM device snapshot (`fake_manila`); the calibrated curve sits *between* the ideal and the harsh uniform toy, i.e. the uniform assumption overstates the damage. Code: `src/block4.py`, plots 8 and 9.

## 1 Setup: what was idealized

The Block-4 pipeline prepares the even-parity ground state with a parity-penalized VQE cost

$$C(\theta) = \langle H \rangle_\theta + \lambda(1 - \langle \hat{P} \rangle_\theta)^2, \quad (1)$$

and then reads the non-local edge string  $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$  on the prepared state. Two idealizations entered:

1. **Noiseless preparation.**  $C(\theta)$  was evaluated on the pure statevector; the optimizer never saw noise and returned the ideal optimum  $\theta^*$ . Noise was applied only afterward, to the fixed  $\theta^*$ .
2. **Toy noise.** The channel was a single uniform per-cx depolarizing probability  $p_{cx}$ , or one global thermal  $T_1/T_2$  applied identically to every qubit.

Sections 2 and 3 remove these one at a time.

## 2 Noisy VQE optimization

### 2.1 Method

We replace the statevector cost by its noisy counterpart: the transpiled preparation circuit is run as a density matrix through a Qiskit `NoiseModel`, and

$$C_{\text{noisy}}(\theta) = \text{Tr}[H \rho_\theta] + \lambda(1 - \text{Tr}[\hat{P} \rho_\theta])^2, \quad \rho_\theta = \mathcal{E}(U(\theta)|0\rangle\langle 0|U^\dagger(\theta)), \quad (2)$$

| strength                                                    | noisy $ \langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle $ |              | noiseless fidelity to ED GS |              |
|-------------------------------------------------------------|----------------------------------------------------------|--------------|-----------------------------|--------------|
|                                                             | best-case                                                | noise-aware  | best-case                   | noise-aware  |
| <i>Depolarizing (unital), <math>p_{cx}</math>:</i>          |                                                          |              |                             |              |
| 0.04                                                        | 0.711                                                    | 0.712        | 0.998                       | 1.000        |
| 0.08                                                        | 0.502                                                    | 0.501        | 0.998                       | 1.000        |
| 0.12                                                        | 0.349                                                    | 0.366        | 0.998                       | 1.000        |
| <i>Amplitude damping (non-unital), <math>\gamma</math>:</i> |                                                          |              |                             |              |
| 0.04                                                        | 0.656                                                    | <b>0.755</b> | 0.998                       | 0.999        |
| 0.08                                                        | 0.422                                                    | <b>0.565</b> | 0.998                       | 0.994        |
| 0.12                                                        | 0.266                                                    | <b>0.337</b> | 0.998                       | <b>0.966</b> |

Table 1: Best-case ( $\theta^*$ , noiseless optimum) vs. noise-aware (re-optimized under the channel),  $L = 4$ ,  $r = 3$ ,  $\mu = 0$ ,  $\lambda = 0.1$ , seed 7. Under depolarizing noise both quantities are unchanged. Under amplitude damping the measured edge string *rises* while the noiseless fidelity to the true ground state *falls* at strong damping.

where  $\mathcal{E}$  fires a channel after every **cx**. Each evaluation is one  $4^L$  density-matrix simulation, so the optimizer (COBYLA) is warm-started from the noiseless optimum  $\theta^*$  and refined, with a few random restarts; the best noisy cost is kept. The warm start guarantees the two strategies coincide at zero noise. Everything is at  $L = 4$ ,  $r = 3$ ,  $\mu = 0$  (sweet spot),  $\lambda = 0.1$ .

## 2.2 Unital vs. non-unital noise

The two columns of Figure 1 sweep the *same* per-**cx** error axis with two channels of opposite character:

- **Depolarizing** is *unital*,  $\mathcal{E}(I) = I$ : it contracts all traceless expectations toward the maximally mixed state without singling out a direction. Empirically the noise-aware optimum is indistinguishable from  $\theta^*$  in every observable (Table 1, top): there is nothing for re-optimization to exploit.
- **Amplitude damping** is *non-unital*: it drives every qubit toward  $|0\rangle$ , breaking the symmetry. The energy optimum genuinely moves, and the noisy optimizer finds a different  $\theta$ .

## 2.3 The fidelity honesty check

A lower noisy energy or a higher measured edge string is meaningless if the underlying state has drifted *away* from the true Majorana ground state – the optimizer would then be exploiting the channel rather than preparing the state better. The decisive quantity is therefore the **noiseless** fidelity  $|\langle \psi_{\text{ED}} | \psi(\theta) \rangle|^2$  of the re-optimized parameters (Table 1, last columns; Figure 1, bottom row).

The result is unambiguous. At *weak* damping ( $\gamma = 0.04$ ) the noise-aware state keeps essentially the same fidelity ( $0.998 \rightarrow 0.999$ ) while raising the edge string – genuine variational compensation. At *strong* damping ( $\gamma \gtrsim 0.06$ ) the edge string still rises, but the noiseless fidelity *drops* (to 0.94–0.97) and does so non-monotonically and seed-dependently: the “recovery” is partly the optimizer gaming the observable through the channel, not a better-prepared ground state. So the honest headline is not “noise-aware VQE recovers the signal” but:

*minimizing the device-level cost is not the same as preparing the ground state.*

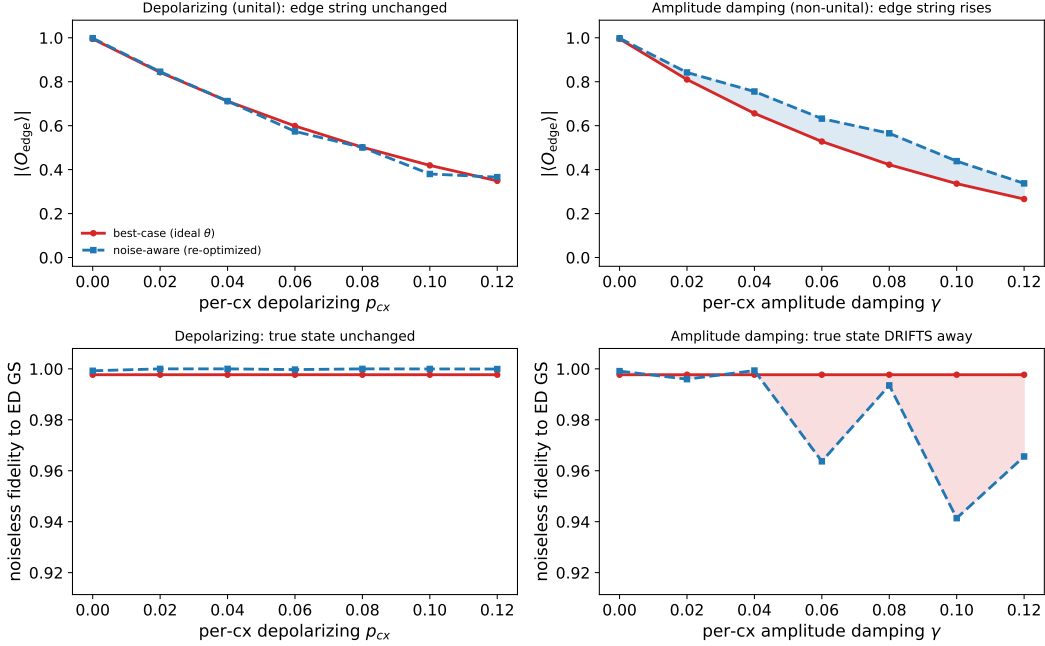


Figure 1: Best-case vs. noise-aware optimization. Columns: depolarizing (unital) | amplitude damping (non-unital), same per-cx axis. Top: the *measured* noisy edge string; bottom: the *noiseless* fidelity to the ED ground state. Non-unital noise lets the optimizer inflate the top-row observable while the bottom-row fidelity falls.

Conversely, the unital null is itself a useful result: it shows that for symmetric gate noise the noiseless-preparation idealization of the rest of Block 4 loses nothing – the best-case curves are representative, not optimistic.

### 3 Backend-calibrated noise

#### 3.1 Method

`NoiseModel.from_backend(backend)` builds a channel from a real device’s recorded calibration – per-qubit  $T_1/T_2$ , per-gate error rates, gate durations, and readout error – rather than one hand-set number. We use the IBM `fake_manila` snapshot (a five-qubit linear device shipped with `qiskit_ibm_runtime`); its `cx` error rates are heterogeneous,  $\approx 0.009$ – $0.014$ , and its linear coupling matches the chain, so every nearest-neighbour `cx` of the ansatz carries a calibrated error. The fixed VQE angles are then evaluated under this model and compared with the uniform toy on the identical circuit.

#### 3.2 Result

At the sweet spot the calibrated model costs a meaningful  $\sim 12\%$  of the edge signal, but far less than the deliberately harsh uniform toy: because the real per-cx rates ( $\sim 0.01$ ) are roughly five times smaller than the toy’s  $0.05$ , a uniform toy model *overstates* the damage (Table 2, Figure 2). The

| $ \langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle $ at $\mu = 0$ | ideal | backend ( <code>fake_manila</code> ) | uniform toy ( $p_{cx} = 0.05$ ) |
|-----------------------------------------------------------------|-------|--------------------------------------|---------------------------------|
| value                                                           | 0.99  | 0.87                                 | 0.67                            |

Table 2: The calibrated device curve sits between the ideal and the uniform toy.

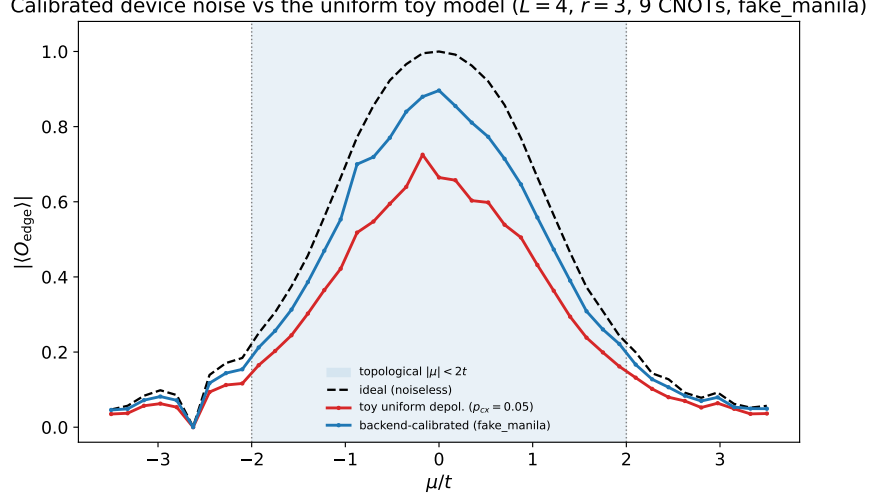


Figure 2: Edge string across the phase diagram under ideal, uniform-toy, and backend-calibrated (`fake_manila`) noise. The calibrated curve lies between the other two.

dominant driver is the per-gate error *magnitude*; per-qubit heterogeneity is a secondary refinement at  $L = 4$ .

### 3.3 Caveats

- **Readout error is excluded.** The diagnostic is read as  $\text{Tr}[X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rho]$  directly from the saved density matrix, with no measurement instruction in the circuit. `from_backend` does include a `measure` error, but it never fires on this path – so even the device curve is mildly optimistic, since real readout adds 1–3% per qubit.
- **Snapshot, not live hardware.** `fake_manila` is recorded calibration data, representative of the device at snapshot time, not a live run. It requires `qiskit_ibm_runtime`; absent that, the code falls back to a synthetic `GenericBackendV2`, whose (seed-generated) error rates are much smaller and should *not* be read as a real-device result.

## 4 Takeaways

- Optimizing a noisy cost does not by itself buy a better *state*. For symmetric (unital) gate noise it changes nothing; for non-unital noise it can raise a chosen observable while degrading fidelity to the target – an artifact to be reported honestly, not a mitigation.
- The unital null result is the quiet payoff: it certifies that the noiseless-preparation idealization used elsewhere in Block 4 is not hiding an optimistic bias for depolarizing noise.

- Uniform toy noise is a conservative caricature: a calibrated device model is milder, though still not noise-immune, and the device curve here is itself optimistic because readout error is excluded.

### **Reproduce.**

```
python src/block4.py --plots 8    # noisy VQE optimization (unital vs non-unital)
python src/block4.py --plots 9    # toy vs backend-calibrated noise
```

Plot 8 uses `amp_damp_noise_model` / `noisy_optimize`; plot 9 uses `backend_noise_model` (`NoiseModel.from_backend`).